UNIVERSITÀ di VERONA

Deep Learning for OCT-Based Classification of Visual Acuity in **Wet AMD**

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Outline

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- Objectives
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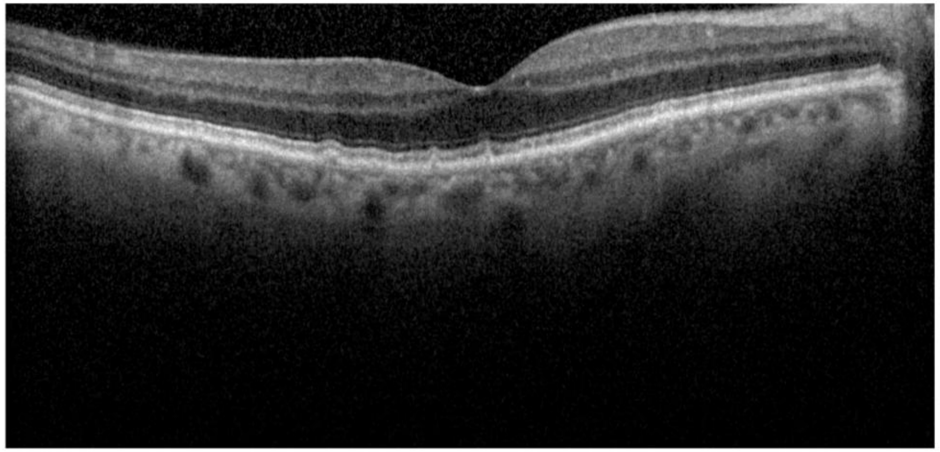
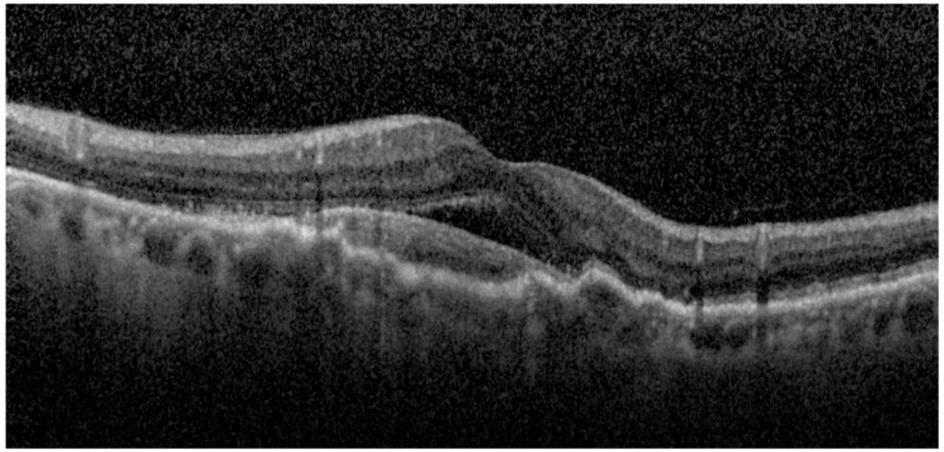
- Data & Labels (dataset)
- Methods: CNN and LSTM models
- Results: Static and Temporal tasks
- Interpretability (Grad-CAM)
- Limitations & Future Work



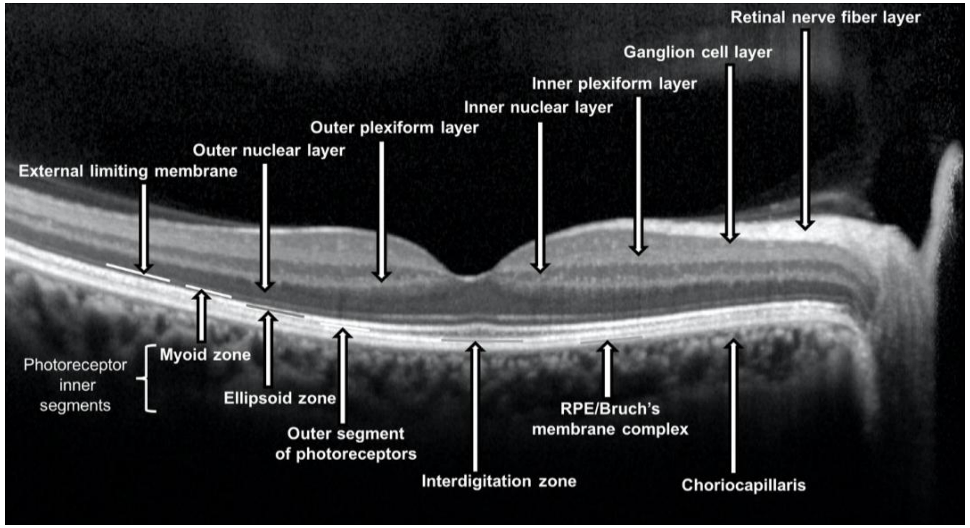


AMD Overview and Clinical Motivation

Age-related macular degeneration (AMD) is a progressive retinal disease and a leading cause of central vision loss in older adults. AMD has two forms: **dry (non-neovascular)** (~90% of cases; gradual vision loss) and **wet (neovascular)** AMD, characterized by abnormal blood vessels under the macula that leak fluid, causing rapid central vision decline. Wet AMD (wAMD) requires frequent monitoring (e.g. anti-VEGF therapy), yet remains a major cause of severe impairment. Its prevalence is growing (projected ~288 million cases by 2040). Accurate prediction of visual outcomes (e.g. “will the patient maintain good vision?”) could help personalize treatment plans and follow-up.

Optical Coherence Tomography (OCT) in AMD

Optical Coherence Tomography (OCT) is a non-invasive, high-resolution imaging modality that provides cross-sectional “B-scan” images of the retina. In AMD care, OCT is indispensable: it reveals retinal layer structure, fluid pockets and lesions that underlie visual acuity changes. Typical wAMD management involves obtaining an OCT scan at each visit to guide therapy (for example, deciding on anti-VEGF injections). The OCT image here illustrates the layered retina in practice; clinicians look for fluid or disruptions in the foveal (macular) region that correlate with vision loss.



Objectives

1. **Static classification** – Develop a deep learning model that takes OCT images of a year and classifies the vision outcome, which can be ‘Good’ or 'Poor’.
2. **Explainability and interpretation (for static classification)** – Implement and evaluate explainability techniques (e.g., Grad-CAM) to highlight retinal features influencing model predictions.
3. **Temporal classification** – Develop a deep learning approach that analyzes a sequence of OCT images from the same eye over multiple visits (a time series spanning months or years) to predict or classify the patient's visual outcome. This sequence-based prediction leverages the temporal evolution of the disease, potentially capturing trends (improvement or deterioration) that a single snapshot cannot.

AI in Ophthalmology – State of the Art

In recent years, AI and deep learning have started to revolutionize medical diagnostics, including applications in ophthalmology. Deep learning, particularly **convolutional neural networks (CNNs)**, has achieved expert-level performance in detecting diseases from medical images. In ophthalmic diagnostics, deep learning systems have demonstrated high accuracy in identifying conditions like diabetic retinopathy and AMD from retinal photographs and OCT scans.

Examples:

- A deep learning model developed by researchers at **Google DeepMind** was able to analyze 3D OCT scans and recommend referral decisions for over 50 eye diseases with about **94% accuracy** – on par with expert clinicians.
- Romo-Bucheli-2020, tackle a **temporal prediction** task in wet AMD based on **anti-VEGF injections**, predicting **injection need** in a patient.

Dataset

- **Data Source:** Longitudinal dataset of **271 wAMD patients (321 eyes)** collected at a single center (Ospedale Sacro Cuore – Don Calabria, located in Negrar, Italy), with follow-up of 8–14 years per patient.
- **File Conversion** : The original scan data was available only through the proprietary Heidelberg Engineering software in the **'.e2e'** format, which encapsulates 3D volumetric OCT data along with fundus images and metadata. Due to the constraints in the exportation system, data could not be retrieved in standard formats such as DICOM or PNG directly in bulk. As the number of patients were larger, manual downloading of data was not feasible. To overcome this, the open-source Python library OCT-Converter (*https://github.com/marksgraham/OCT-Converter*) was used.
- **Preprocessing:** B-scans were quality-filtered and standardized. OCT images were resized to 224×224 and converted to single-channel (grayscale) for input. Images were grouped by year of follow-up for each eye, and eyes were separated by laterality. Each year is paired with best-corrected visual acuity (BCVA) and treatment records.

Labeling

- **Static Labeling (per year):** Each year's scans are labeled

$$Label_{year} = \begin{cases} \boldsymbol{good}, & if\ BCAV_{year} \geq 0.4 \\ \boldsymbol{bad}, & otherwise \end{cases}$$

- **Temporal Labeling:** For predicting year-over-year change,

$$\Delta BCVA_n = BCVA_n - BCVA_{n-1}$$

The labeling is as follows,

$$Label_n = \begin{cases} \boldsymbol{baseline}, & if\ n = 1 \\ \boldsymbol{improved}, & if\ \Delta BCVA_n \geq 0.1 \\ \boldsymbol{not-improved}, & if \Delta BCVA_n < 0.1 \end{cases}$$

Static Classification Models - (CNN Architectures)

Fine-tuned three ImageNet-pretrained CNNs adapted for single-channel OCT:

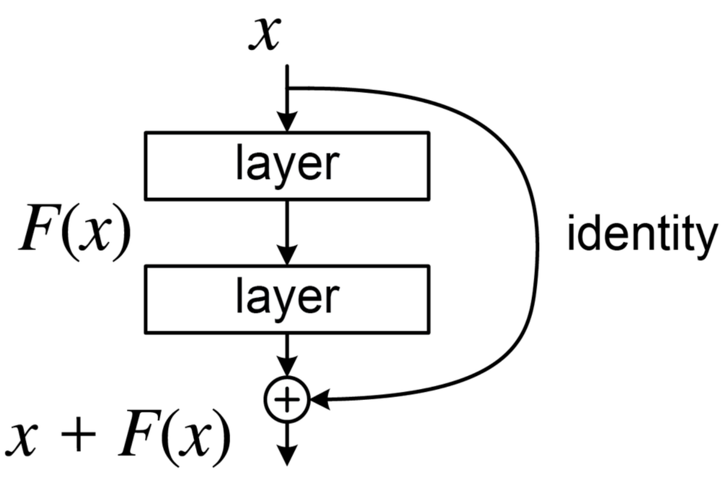
ResNet18,

ResNet34, and

DenseNet121.

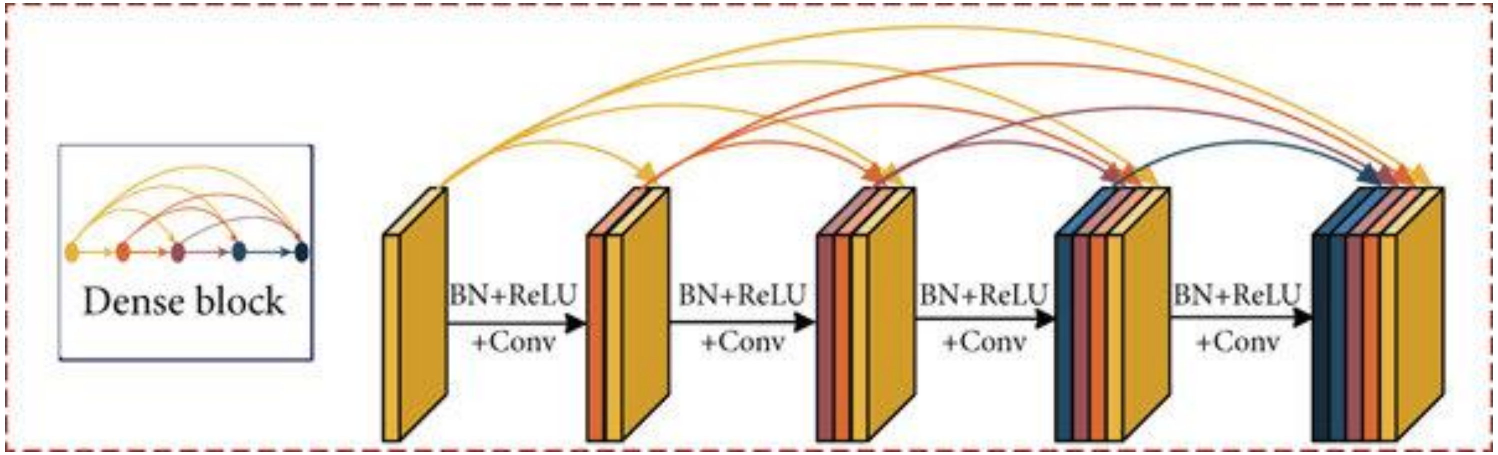
These were chosen for their proven performance in medical imaging and differing depth/connectivity.

ResNet (Residual Network): Uses identity skip-connections to train deep nets without vanishing gradients. ResNet18 and 34 are 18- and 34-layer variants that are relatively lightweight and effective on smaller datasets.



A residual block used in ResNet architecture. The input x is forwarded (identity shortcut) and added to the output of a sequence of layers computing F(x) (e.g., Conv+BN+ReLU). This skip connection enables the network to learn residual functions (F(x)) on top of the identity, which greatly eases training of deep networks by mitigating vanishing gradients. ResNet34 stacks such residual blocks (with occasional downsampling layers) to form a deep CNN for image classification

DenseNet: DenseNet121 (121 layers) uses “dense” connectivity (each layer feeds all subsequent layers) to encourage feature reuse. DenseNets often yield richer feature representations, which is beneficial for detailed OCT images.



DenseNet architecture and dense block connectivity. Each dense block (yellow box) consists of multiple convolutional layers (dense units) where each layer takes all preceding feature-maps as input (illustrated by the multiple arrows). The feature-maps are concatenated channel-wise before feeding into each new layer’s BN–ReLU–Conv operations. This dense connectivity pattern enables maximum feature reuse and strengthens gradient f low through the network

Temporal Classification Model - (ResNet + LSTM)

- **Task:** Predict year-over-year **improvement** in visual acuity (BCVA gain ≥ 0.1) using sequential OCT features.
- **Hybrid Architecture:** Each annual OCT scan is encoded by a shared ResNet18 (up to the global pooling layer) into a 512-dimensional feature vector. For a patient with *T* years of follow-up, this yields a sequence of vectors $[f_1, \ldots, f_T]$.
- **LSTM Sequence Model:** The sequence of ResNet features is fed into a 2-layer LSTM (hidden size 128). We predict "improved" vs "not improved" for each year by taking the LSTM's final output (after the last year's features) through a sigmoid classifier.

Experiments

- **Code Availability:** *https://github.com/JawwadMehdi14/oct-visual-acuity-wet-amd*
- **Static classification:**
  - We treat '**good vision**' as the positive class (i.e., a true positive is a correctly identified good vision exam).
  - Evaluated three deep CNN architectures: ResNet18, ResNet34 and DenseNet121.
  - Fine-tuned each model on our dataset using the Adam optimizer (initial learning rate $1 \times 10{-}4$) and binary cross-entropy loss for the binary classification ("good" vs. "poor" vision).
  - Each training was run for 30 epochs per fold with a mini-batch size of 32.
  - Model selection was based on validation performance (retained the epoch with the highest validation F1-score to avoid overfitting).
  - Performed a 3-fold cross-validation in all experiments.
  - Compared two split strategies: (1) **image-wise splitting**, (2) **patients' eye level splitting**.

Experiments

- **Temporal classification:**
  - Focused on the '**not improved**' class as positive.
  - A hybrid deep learning model that combines a CNN for feature extraction (ResNET18) and an LSTM.
  - Adam optimizer with an initial learning rate of $1\times10{-}3$ and the Focal Loss ($\gamma = 2$) as our loss function to address class imbalance.
  - Each training was run for 30 epochs per fold with a mini-batch size of 32.
  - Model selection was based on validation performance (retained the epoch with the highest validation F1-score to avoid overfitting).
  - Performed a 3-fold cross-validation in all experiments.
  - Compared two strategies: (1) **full history**, (2) **last-year only**.

Static Classification Results

Metrics	**ResNet18**	**ResNet34**	**DenseNet121**
Accuracy	0.8168 ± 0.0308	0.7945 ± 0.0328	0.8372 ± 0.0088
Precision	0.7490 ± 0.0475	0.7300 ± 0.0479	0.7881 ± 0.0190
Recall	0.8984 ± 0.0166	0.8688 ± 0.0362	0.8742 ± 0.0100
F_1-score	0.8160 ± 0.0220	0.7924 ± 0.0282	0.8287 ± 0.0062
AUC	0.9200 ± 0.0093	0.8966 ± 0.0258	0.9234 ± 0.0024

Image-wise splitting

Explainability Analysis with Grad-CAM

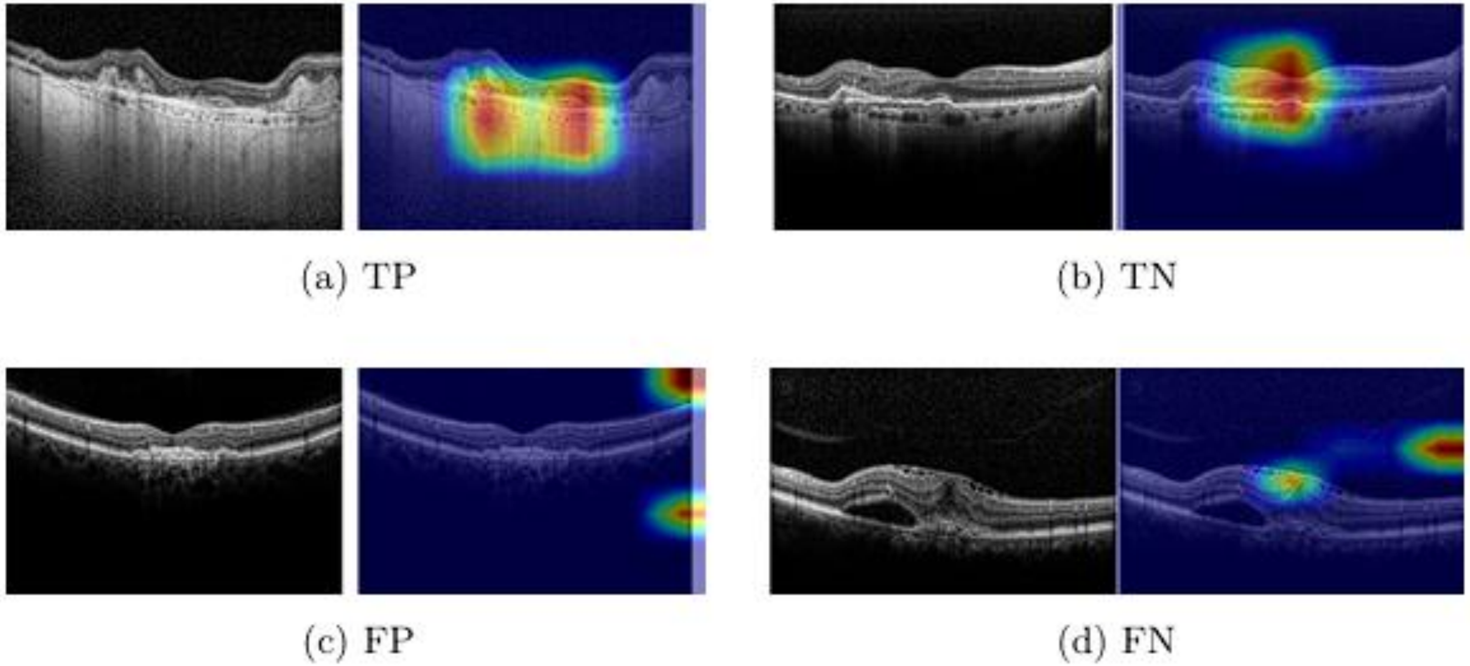


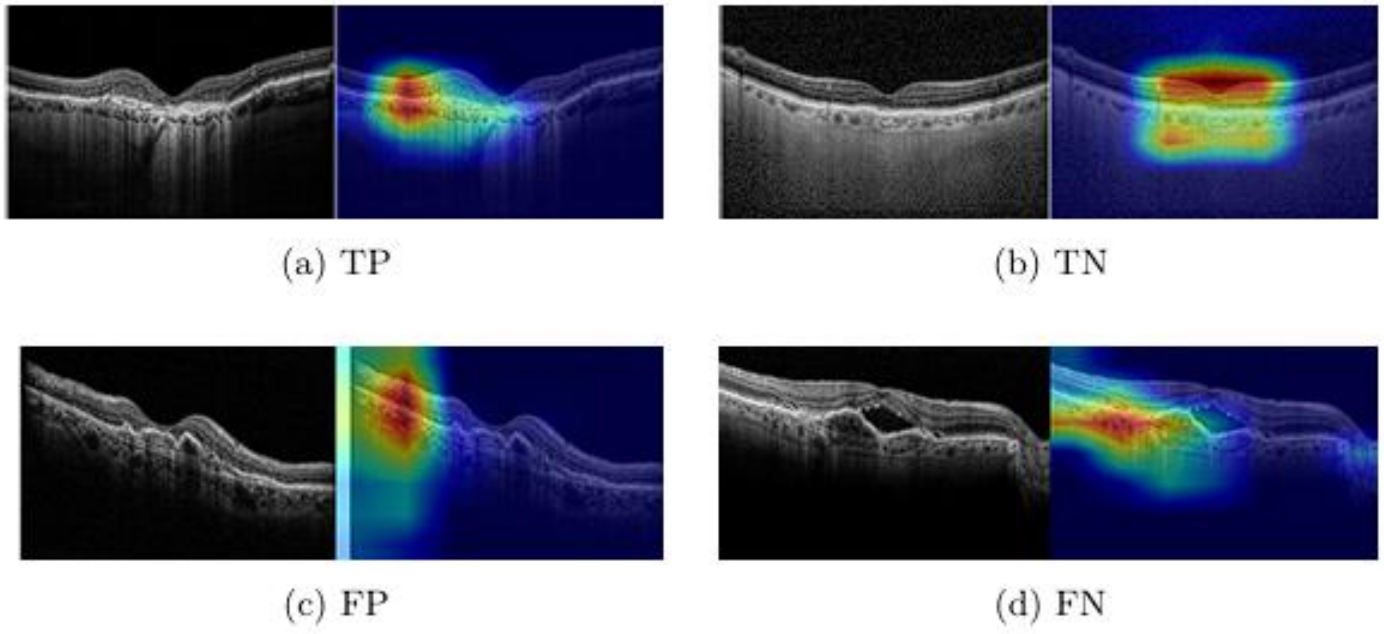
Image-wise explainability results showing (a) True Positive, (b) True Negative, (c) False Positive, and (d) False Negative for image-wise splitting.

Static Classification Results

Metrics	**ResNet18**	**ResNet34**	**DenseNet121**
Accuracy	0.6891 ± 0.008	0.6757 ± 0.0175	0.6932 ± 0.0301
Precision	0.7138 ± 0.0272	0.6864 ± 0.0507	0.7116 ± 0.0473
Recall	0.7960 ± 0.0257	0.8471 ± 0.0647	0.8164 ± 0.0344
F_1-score	0.7521 ± 0.0074	0.7556 ± 0.0092	0.7592 ± 0.0233
AUC	0.7615 ± 0.0157	0.7661 ± 0.0115	0.7664 ± 0.0209

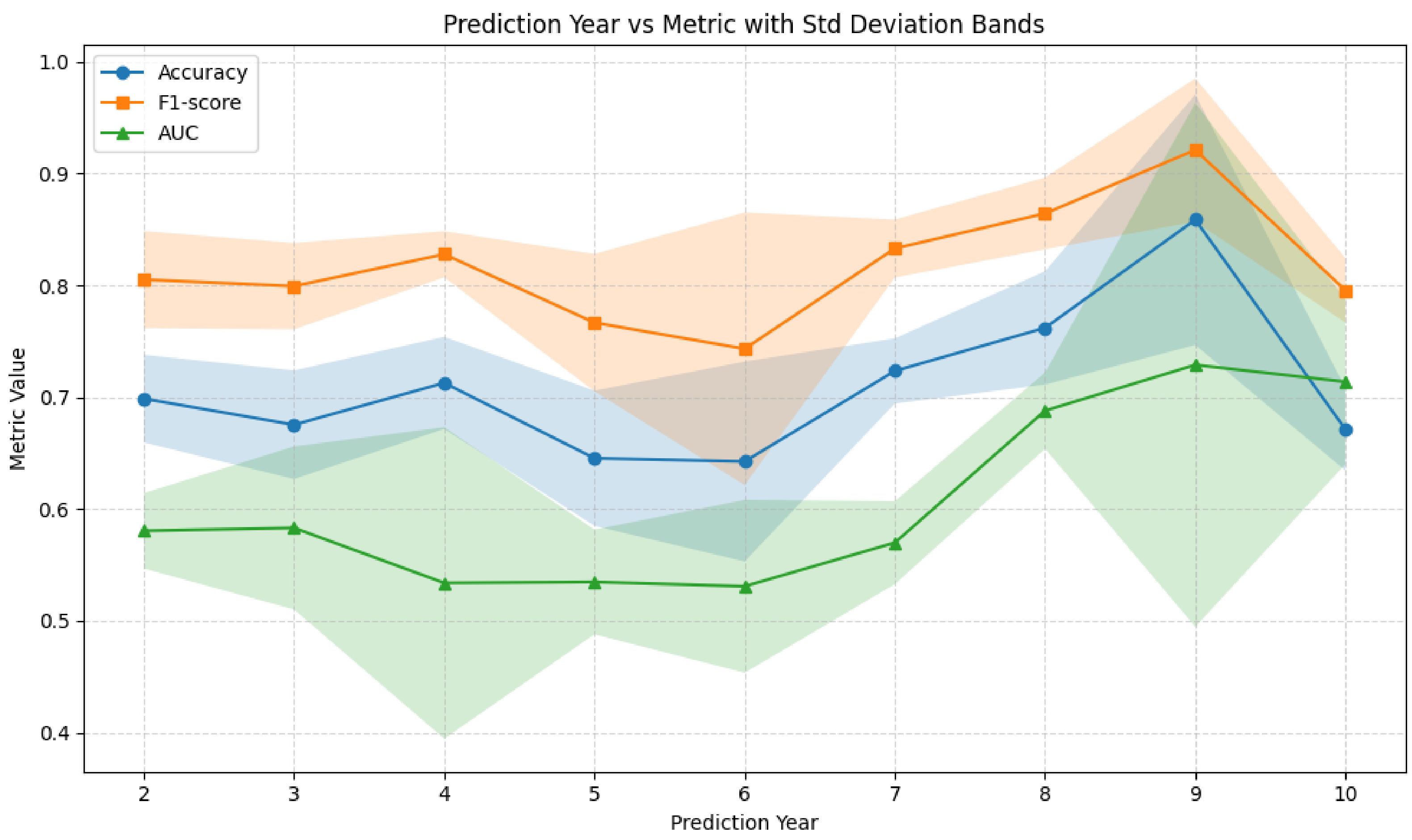
Patient-wise splitting

Explainability Analysis with Grad-CAM



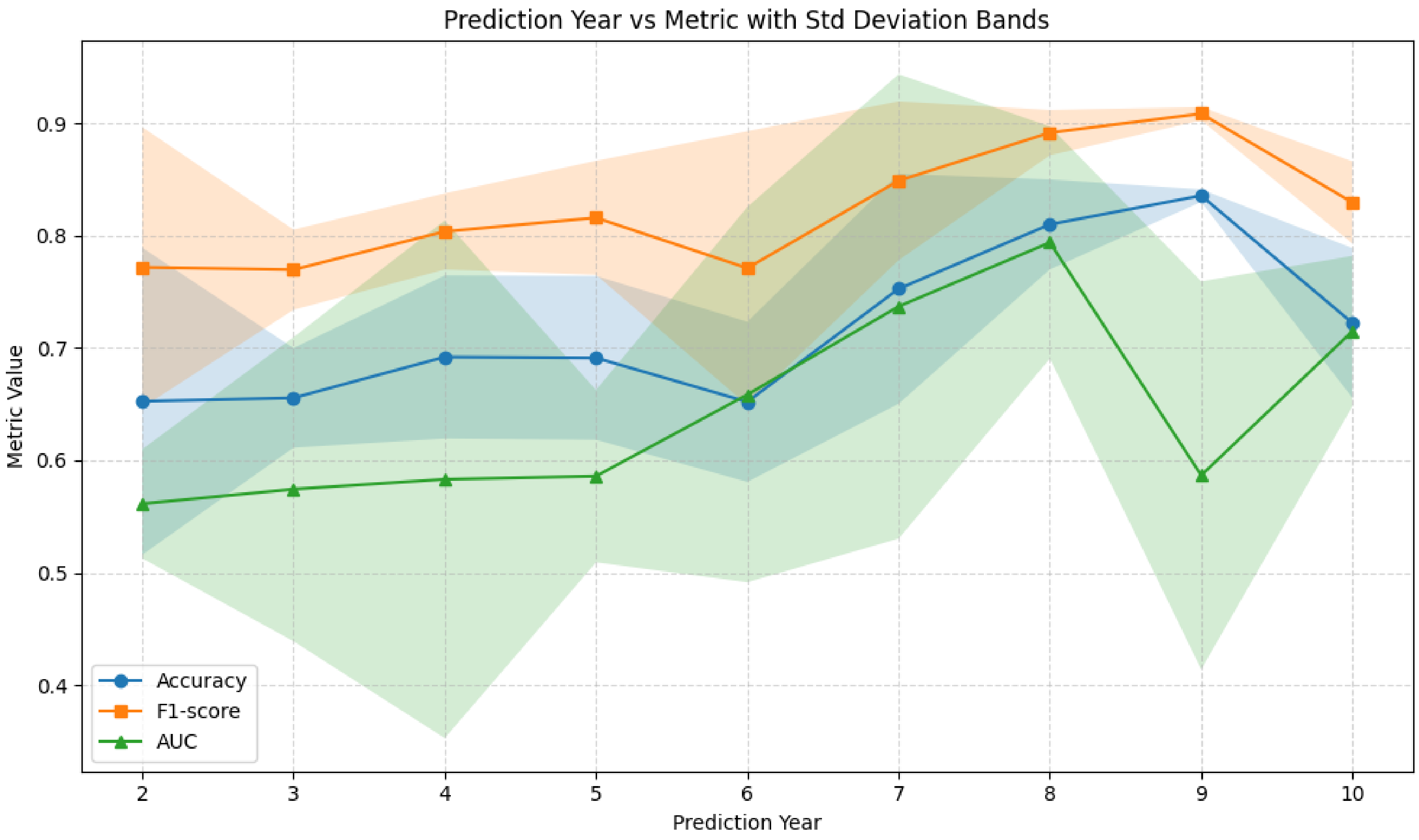
Patient-wise explainability results showing (a) True Positive, (b) True Negative, (c) False Positive, and (d) False Negative for image-wise splitting.

Temporal Classification Results



Temporal trend (based on full history)

Temporal Classification Results



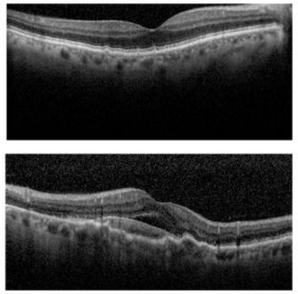
Temporal trend (based on last year)

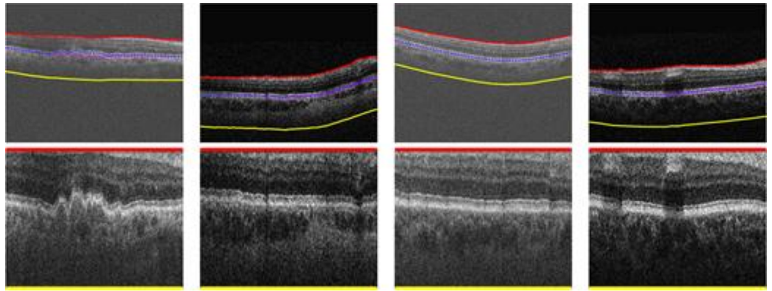
Limitations

- **Data Imbalance:** Especially in the temporal task – was a significant challenge. In many follow-up years, far fewer patients were labeled “improved” than “not improved”, making it hard for the model to learn the minority class.
- **Patient Variability:** The progression of wet AMD and response to therapy vary greatly between individuals.
- **Yearly Grouping of OCT Scans:** Grouped OCT scans and clinical measures by year. This was done due to time constraints and because the clinical outcomes (BCVA changes) were easiest to quantify annually. However, in reality scans were taken more frequently (approximately every 2 months).
- **Pre-processing:** Pipeline used the raw OCT B-scans (after basic re sizing and filtering) without heavy pre-processing.
- **Temporal Modeling Strategy:** LSTM model predicts one year ahead at a time, effectively chaining predictions, this sequential prediction approach could compound errors — a mistake in interpreting an earlier year might carry forward.

Future Works

- **Multimodal data integration:** Future models could combine OCT images with other data sources - for example, color fundus photographs, visual acuity history, or biomarkers from electronic health records.
- **Advanced architectures and pre-processing:** Future research could explore more advanced deep learning architectures. Transformer-based models or attention mechanisms, for in stance, might capture long-range dependencies in a sequence of OCT scans more effectively. Additionally, more systematic pre-processing—such as retinal layer segmentation and standardized cropping—could help models focus on clinically meaningful features.





- **Advanced forecasting strategies:** for example, models that predict multiple future time points jointly rather than one-step-at-a-time.

Thank you for your time and attention.